

WILLIAM R. JOHNSON

SOFTWARE ENGINEER

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Atlanta, GA
CooCooCaCha

EXPERIENCE

CancerIQ
2022 - present

Principal Software Engineer
Promoted from Senior Software Engineer in 2026.

Built apps for cancer prevention and early detection using Rails, React, Rust, Elixir, Node.js, and Google Cloud.

- Led the genogram team in rewriting our visual family tree builder in React and refactoring the Rails backend to support new features.
- Researched and developed a deterministic, recursive family tree layout algorithm that preserves paternal/maternal orientation and keeps twins adjacent, replacing our previous Sugiyama-based layout.
- Built a custom query language using parser combinators, enabling product managers to target content based on patient data.
- Worked with a co-founder to standardize an Excel format for patient data and built a Ruby import pipeline using pre-generated UUIDs and transactions to prevent partial patient imports and resume safely after failures. Tested with 1M+ records and 100+ columns.
- Led improvements to our AI-assisted development workflow by grouping apps and services in one folder and writing an AGENTS.md to give AI tools codebase context and development guidelines.

World 50
2018 - 2022

Principal Software Engineer
Promoted from Senior Software Engineer in 2021.

Full-stack engineer and manager of five engineers working on the company's React Native and Node.js platform.

- Rebuilt the legacy application with React Native, sharing most of the code across web, iOS, and Android.
- Rebuilt the API with GraphQL, using DataLoader to avoid N+1 queries.
- Proposed an engineering process, later adopted, in which temporary, cross-functional teams planned and delivered individual epics.
- Set up databases, servers, and build pipelines using AWS, Google Cloud, and CircleCI.
- Introduced architecture decision records to document engineering decisions.

TSYS
2017 - 2018

Senior Software Engineer
ReactJS and front-end specialist.

- Gave presentations and introduced new React techniques such as Fragments and Render-Props (hooks were released after I left, but we would have used those instead).
- Completely rebuilt the UI for my product.
- Built a dynamically generated page flow system. Multi-page flows were generated from data returned from our REST API.
- Regularly contributed to our core React frameworks.
- Mentored and trained junior developers on React.

Koble
2016 - 2017

Co-Founder / Lead Engineer
Made all of the software architecture decisions. Wrote all of the code. Responded to outages and server issues.

- Developed a homepage using React and a prototype app using React-Native.
- Setup and maintained infrastructure using Google Cloud, including a Kubernetes cluster.
- Built a service to automatically retrieve and update LetsEncrypt certificates for TLS/SSL encryption.
- Wrote an audit-log service that listened for changes in Firebase and inserted a change summary into Google BigQuery.

Salesloft
2015 - 2016

Full-Stack Software Engineer
Designed, developed, and deployed features for the Salesloft SAAS application. This touched upon everything from database design and migrations to front-end UX design and development.

- Developed a new QA system using Docker. Different versions of our application could be tested simultaneously on the same server cluster.
- Learned Ruby, Ruby on Rails, and Angular on the job. Started contributing quickly.

EDUCATION

Georgia Institute of Technology
Bachelor's Degree - Computer Science
2007 - 2012

SKILLS

Javascript	Typecript	React	React-Native
GraphQL	Rust	NodeJS	SQL
Git	AWS and Google Cloud	HTML	CSS
Rails	Python	Ruby	